Periodic Table

Subject: Chemistry | NoteShare Library

PERIODIC TABLE

1. STRUCTURE
7 periods 18 groups
s, p, d, f blocks

2. GROUPS
Group 1: Alkali, +1
Group 2: Alkaline earth, +2
Group 17: Halogens, -1
Group 18: Noble, inert

3. TRENDS
Property | L->R | T->B
Atomic radius | Decreases | Increases
Ionization E | Increases | Decreases
Electron affinity| Increases | Decreases
Electronegativity| Increases | Decreases
Metallic char | Decreases | Increases

4. ELECTRON CONFIG
Aufbau: lowest first
Pauli: max 2 per orbital
Hund: single fill first

Order: 1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 4f 5d 6p

Examples:
H: 1s
C: 1s 2s 2p
Na: [Ne] 3s

5. IMPORTANT ELEMENTS
H (most abundant), He, C (organic), N (78% air), O (21%)
Na, K (nerve), Ca (bones), Fe (Hb), Cu (wires)

6. ALLOYS
Steel: Fe+C
Bronze: Cu+Sn
Brass: Cu+Zn
Stainless: Fe+Cr+Ni